

OIS-007 — Conversation Engine

Status: Foundational

Authority: Implementation

Depends on:

- OIS-000 through OIS-006
 - OAS-012 — Conversation Architecture
 - OCS-000 through OCS-032
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Purpose

This specification defines the implementation responsibilities of the Conversation Engine.

The Constitution defines Conversation as the primary process through which Understanding evolves.

The Architecture defines the organization of Conversation.

This specification defines the executable engine that realizes conversational evolution.

The Conversation Engine SHALL execute Conversations.

It SHALL never define their constitutional meaning.

1. Implementation Objective

The objective of the Conversation Engine is to operationalize constitutional conversations while preserving reasoning, history, attribution, and continuity.

The Engine SHALL preserve:

Conversation identity.

Voice attribution.

Turn ordering.

Context.

Reasoning lineage.

Historical reconstruction.

2. Conversation Lifecycle

Every Conversation SHALL support the following operational lifecycle.

```
graph TD; A[Create] --> B[Initialize Context]; B --> C[Accept Turns]; C --> D[Link Evidence]; D --> E[Record Events]; E --> F[Update Projections]; F --> G[Support Reflection]; G --> H[Continue Evolution];
```

Create

↓

Initialize Context

↓

Accept Turns

↓

Link Evidence

↓

Record Events

↓

Update Projections

↓

Support Reflection

↓

Continue Evolution

No lifecycle transition SHALL erase historical continuity.

3. Conversation Context

The Engine SHALL maintain operational context including:

Active Place.

Applicable Schema.

Participating Voices.

Relevant Questions.

Referenced Evidence.

Related Conversations.

Current Understanding projections.

Context SHALL remain reconstructable.

4. Conversation Turns

Every conversational turn SHALL generate one or more immutable Events.

Each turn SHALL preserve:

Identity.

Voice.

Intent.

Context.

Relationships.

Operational metadata.

Operational metadata SHALL remain implementation-specific.

Constitutional meaning SHALL remain unchanged.

5. Reasoning Support

The Engine SHALL support explicit reasoning relationships.

Implementation SHALL preserve links between:

Questions.

Evidence.

Arguments.

Counterarguments.

Explanations.

Emerging Understanding.

Reasoning SHALL remain navigable.

6. Projection Services

The Engine MAY maintain operational projections such as:

Conversation summaries.

Current topics.

Open Questions.

Participation views.

Learning indicators.

All projections SHALL remain reconstructable from constitutional history.

7. Federation

Conversation Engines MAY exchange constitutional Conversations.

Federation SHALL preserve:

Canonical identity.

Voice attribution.

Historical ordering.

Relationship integrity.

Local ownership.

Conversation federation SHALL remain asynchronous.

8. Recovery

Recovery SHALL reconstruct Conversations using:

Identity.

Event history.

Schema.

Architectural relationships.

Referenced constitutional constructs.

Recovery SHALL preserve conversational continuity.

9. Verification

Verification SHALL demonstrate:

Correct ordering.

Identity continuity.

Reasoning preservation.

Projection equivalence.

Recovery correctness.

Equivalent constitutional conversation behavior.

10. Conversation Pattern

The Conversation Engine SHALL optimize execution.

It SHALL NOT optimize away reasoning, attribution, uncertainty, or historical continuity.

Operational efficiency SHALL remain subordinate to constitutional understanding.

11. Conversation Engine Test

A Conversation Engine satisfies this specification when independent implementations can reconstruct constitutionally equivalent Conversations, including reasoning, participation, uncertainty, and historical evolution, despite differences in infrastructure, storage, execution models, or user interfaces.

Conformance

An implementation conforms to this specification when Conversations remain operationally executable, historically continuous, constitutionally attributable, and architecturally equivalent independently of implementation technology.

Implementation Principle

Most systems treat conversations as communication.

Organodynamics treats conversations as the operational engine of understanding.

The Conversation Engine therefore exists not merely to exchange messages,

but to preserve the complete constitutional process through which Questions become reasoning,

reasoning becomes understanding,

and understanding becomes the foundation for future constitutional evolution.

End of Specification OIS-007